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Analysis of magnetohydrodynamics oscillations at Heliotron J with Python

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AI Usage Declaration

During this work, the AI models Claude Sonnet 5, Claude Opus 4.6, Gemini 3.1 Pro and Gemini 3.7 & 3.8 Flash were used for the following purposes:

- Reviewing spelling, grammar, syntax, translation doubts, and improving the readability and flow of the text.
- Debugging, fixing bugs, and performance optimization of the python codes.

All scientific analyses, computational implementations, and conclusions represent the original work of the author. The author reviewed and verified all AI-generated suggestions and assumes full responsibility for the final contents of this document, as they were checked and discussed with the Heliotron J team.

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1 Introduction

1.1 Identification

1.1.1 Institution

Located at Gokasho, Uji, Kyoto 611-0011, Japan, the Heliotron J project is a research initiative of the Institute of Advanced Energy at Kyoto University, which focuses on the advancement of controlled fusion. This experimental facility utilizes a helical-axis heliotron configuration to investigate the magnetic confinement and contribute to understanding how to achieve the stability of high-temperature plasmas. By optimizing magnetic field geometry and advanced heating methods, the project aims to establish the scientific and engineering foundations required for the development of a sustainable, fusion-based power generation system.

This institute was founded in 1996 by the consolidation of the Institute of Atomic Energy and the Plasma Physics Laboratory, but its history traces back to 1941 and with its first fusion device “Heliotron A”. Since then, the heliotron devices count has reached 8 (A, B, C, D, DM, E, DR and J), the last one, being the center of this research, was built on 1999, celebrating 27 years of existence until today [1]. This institute is also host to several other laboratories and research groups, such as the Advanced Energy Generation Division, Advanced Energy Conversion Division, Advanced Energy Utilization Division, Laboratory for Complex Energy Processes and Integrated Research Center for Carbon Negative Science, resulting in a multidisciplinary and a multilayered institution [2].

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1.2 Context of the work

A major obstacle to securing the obtainment of energy via nuclear fusion are the magnetohydrodynamic (MHD) instabilities [3]. In helical devices, MHD oscillations driven by energetic electrons and/or ions, lead to significant transport and loss of high-energy particles. These processes reduce heating efficiency and energy confinement, producing serious risk of localized damage to plasma-facing components due to particle hot spots [4]. Recent experimental observations in Heliotron J reveal that low-frequency electromagnetic fluctuations are excited during the H-mode phase. These fluctuations couple strongly with edge turbulence, enhancing cross-field heat transport and thereby limiting the expected improvement in energy confinement [5]. Such behavior is considered a key factor underlying the H-mode low performance in helical devices compared to tokamaks. In addition, the excitation of large-amplitude MHD oscillations has been observed under certain operational conditions, highlighting the complexity and variability of plasma stability in these systems [6]. These findings underscore

the urgent need for a comprehensive understanding of both the characteristics and driving mechanisms of MHD oscillations in order to achieve stable plasma operation.

This thesis focuses on energetic particle-driven modes [4] and low-frequency H-mode fluctuations [5], using Mirnov coil, diamagnetic loop, and ECE diagnostics on a curated subset of discharges with consistent magnetic topologies and heating schemes; other instability classes, such as tearing modes or edge-localized modes, fall outside this scope. Multi-channel magnetic fluctuation diagnostics resolve the spatiotemporal structure of MHD activity, while Fourier, Hilbert, and wavelet-based analysis and higher-order correlation techniques extract coherent structures, identifying the mechanisms through which these instabilities limit confinement and how plasma conditions and magnetic configurations trigger MHD activity [6]. It is theorized that these oscillations are determined by specific plasma conditions and magnetic configurations, and that their systematic characterization will identify the dominant excitation mechanisms and their correlation with heating performance, transport, and confinement. The concrete deliverable for the Heliotron J team is a reusable, open-source pipeline, built on the existing jpack library, that produces a validated catalogue of characterized modes as a template for future campaigns.

1.3 Objectives

1.3.1 General Objective

To characterize MHD oscillations in the Heliotron J device through Python-based analysis of experimental data, with the purpose of determining their excitation mechanisms and assessing their impact on plasma confinement.

1.3.2 Specific Objectives

- To characterize the spectral properties and temporal evolution of MHD oscillations in experimental discharges through Python-based signal processing techniques, with the generation of time-frequency representations and the identification of dominant modes.
- To evaluate the correlation between energetic particle modes (EPMs) and plasma heating performance through the analysis of synchronization among magnetic fluctuation frequencies, Neutral Beam Injection (NBI), and Electron Cyclotron Heating (ECH), with the derivation of statistical correlation metrics.
- To identify low-frequency H-mode fluctuations through the analysis of magnetic signals, with the construction of a detailed mapping of nonlinear coupling events associated with edge turbulence modulation and enhanced heat transport.

2 Theoretical framework

2.1 Plasma and Magnetic Confinement Fundamentals

2.1.1 Basic Plasma Parameters and the Confinement Criterion

A common definition of plasma is that it is a quasineutral ionized gas that exhibits collective behavior, requiring the physical system dimensions L to be greater than the Debye length λ_D (m) [7], defined in equation 1:

$$\lambda_D = \sqrt{\frac{\varepsilon_0 k_B T_e}{n_e e^2}} \quad (1)$$

Where ε_0 is the vacuum permittivity (F/m), k_B is the Boltzmann constant (J/K), T_e is the electron temperature (K), n_e is the electron density (m^{-3}), and e is the fundamental charge (C). Quasineutrality, defined in equation 2, applies macroscopically but fails within λ_D , where strong electric fields persist [7]:

$$n_i \simeq n_e \simeq n \quad (2)$$

Where n_i is the ion density (m^{-3}) and n is the plasma density (m^{-3}). Collective plasma behavior is characterized by the electron plasma frequency ω_{pe} (Hz), expressed in equation 3 [7]:

$$\omega_{pe} = \sqrt{\frac{n_e e^2}{m_e \varepsilon_0}} \quad (3)$$

Where m_e is the mass of the electron (kg). In order to obtain a positive fusion energy production, the confinement parameters must satisfy the Lawson criterion given in equation 4, ensuring the fusion reaction liberates more energy than required to sustain it [7]:

$$n\tau_E T \geq 3 \times 10^{21} \text{ m}^{-3} \text{ s keV} \quad (4)$$

Where τ_E is the energy confinement time (s) and T is the plasma temperature (keV). The plasma β parameter measures the ratio of kinetic pressure to magnetic pressure, and it is defined in equation 5 [7]:

$$\beta = \frac{\sum_j n_j k_B T_j}{\frac{B^2}{2\mu_0}} \quad (5)$$

Where j is the species index, n_j is the number density of species j (m^{-3}), T_j is the temperature of species j (K), B is the magnetic field (T), and μ_0 is the vacuum permeability (H/m).

2.1.2 MHD Equations and Equilibrium

The ideal MHD equations describe macroscopic plasma behavior, coupling fluid dynamics to Maxwell's equations through momentum, Ohm's law, state, mass continuity, Ampère's, and

Faraday's laws, presented in equations 6 to 11, respectively [7, 8]:

$$\rho \frac{d\mathbf{u}}{dt} = -\nabla P + \mathbf{J} \times \mathbf{B}, \quad (6)$$

$$\mathbf{E} = -\mathbf{u} \times \mathbf{B}, \quad (7)$$

$$\frac{\partial P}{\partial t} = -\mathbf{u} \cdot \nabla P - \Gamma P \nabla \cdot \mathbf{u}, \quad (8)$$

$$\frac{\partial \rho}{\partial t} = -\mathbf{u} \cdot \nabla \rho - \rho \nabla \cdot \mathbf{u}, \quad (9)$$

$$\mu_0 \mathbf{J} = \nabla \times \mathbf{B}, \quad (10)$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}, \quad (11)$$

Where ρ is the mass density (kg/m^3), $\mathbf{u}$ is the fluid velocity vector (m/s), t is the time (s), P is the thermal pressure (Pa), $\mathbf{J}$ is the current density vector (A/m^2), $\mathbf{B}$ is the magnetic field vector (T), $\mathbf{E}$ is the electric field vector (V/m), and Γ is the adiabatic index. Static MHD equilibrium dictates that nested magnetic flux surfaces support the plasma pressure gradient, formulated in equation 12:

$$\mathbf{J} \times \mathbf{B} = \nabla p \quad (12)$$

Where p is the kinetic plasma pressure (Pa). Perturbations to this equilibrium excite shear Alfvén waves propagating along magnetic flux surfaces [7, 8].

2.2 The Heliotron J Device

Heliotron J is a helical-axis heliotron generated by an $L = 1, M = 4$ continuous helical coil, employing varied toroidal and vertical field coils to produce a rotational transform without a net toroidal current [7, 9, 10]. Heating relies on Neutral Beam Injection (NBI), providing non-Maxwellian energetic ions that serve as the primary drive for kinetic MHD modes, and 70 GHz Electron Cyclotron Heating (ECH) for localized resonance [11]. Diagnostics utilized include: toroidal and poloidal Mirnov coil arrays (MP1–MP4 and PMP1–PMP14) for magnetic fluctuations; diamagnetic loops measuring macroscopic stored energy W_p ; Electron Cyclotron Emission (ECE) for electron temperature profiles; and Langmuir/ H_α FAST arrays for edge characterization [11]. This heating and diagnostic systems locations on the Heliotron J can be observed in the figure 1.

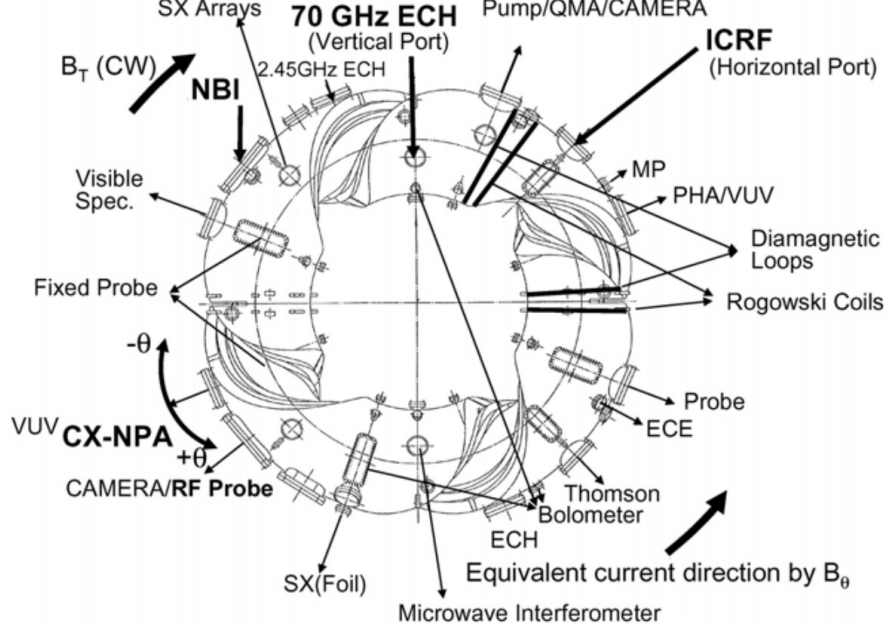


Figure 1: Heating and diagnostic systems in Heliotron J [12].

2.3 MHD Instabilities and Alfvén Eigenmodes

2.3.1 Energetic-Particle-Driven Modes

Energetic particles (EPs) injected by NBI exhibit a bump-on-tail velocity distribution, providing free energy to resonantly destabilize background modes [3, 13, 14]. Shear Alfvén waves propagate at the Alfvén velocity v_A (m/s), defined in equation 13 [7]:

$$v_A = \frac{B}{\sqrt{\mu_0 n_i m_i}} \quad (13)$$

Where m_i is the ion mass (kg). In 3D helical systems, non-axisymmetric geometric couplings open discrete frequency gaps allowing Alfvén eigenmodes (AEs) to exist without heavy continuum damping [3, 14]. Instability requires the wave-particle resonance defined in equation 14 [3]:

$$\omega - n\omega_\phi + L\omega_\theta = 0 \quad (14)$$

Where ω is the mode angular frequency (rad/s), n is the toroidal mode number, ω_ϕ is the energetic particle toroidal transit frequency (rad/s), L is the integer harmonic, and ω_θ is the energetic particle poloidal drift frequency (rad/s). Mode spatial periodicity is defined in equation 15:

$$\exp[i(m\theta - n\zeta)] \quad (15)$$

Where m is the poloidal mode number, θ is the poloidal coordinate (rad), and ζ is the toroidal coordinate (rad). Resonance occurs at the rational flux surface defined in equation 16 [14]:

$$\iota/2\pi = n/m \quad (16)$$

Where ι is the rotational transform (rad).

2.3.2 Theoretical Frequency Scaling Models

To physically classify instabilities, instantaneous frequencies are compared against established scaling models [15]. Shear Alfvén scaling is defined in equation 17:

$$f_A(t) \propto \frac{v_A(t)}{qR_0} \propto \frac{B}{\sqrt{n_e(t)}} \quad (17)$$

Where $f_A(t)$ is the instantaneous shear Alfvén frequency (Hz), q is the safety factor, and R_0 is the major radius (m). Beta-induced Alfvén Eigenmodes (BAE) and Geodesic Acoustic Modes (GAM) scale with temperature as shown in equation 18:

$$f_{\text{BAE}}(t) \propto \sqrt{T_e(t)} \quad (18)$$

Where $f_{\text{BAE}}(t)$ is the instantaneous Beta-induced Alfvén Eigenmode frequency (Hz). Compressibility coupled scaling is given by equation 19:

$$f(t) \propto \sqrt{\frac{C_1}{n_e(t)} + C_2 T_e(t)} \quad (19)$$

Where $f(t)$ is the compressibility-coupled mode frequency (Hz), and C_1 and C_2 are scaling constants. Beam-driven Energetic Particle Modes (EPMs) scale via density gradients or drift resonance, presented in equations 20 and 21:

$$f_{\text{EPM}}(t) \propto \sqrt{\frac{P_{\text{NBI}}(t)}{n_e(t)}} \quad (20)$$

$$f_{\text{EPM}}(t) \propto P_{\text{NBI}}(t) \quad (21)$$

Where $f_{\text{EPM}}(t)$ is the Energetic Particle Mode frequency (Hz) and $P_{\text{NBI}}(t)$ is the Neutral Beam Injection heating power (W). Macroscopic mode shifts due to plasma current and finite- β energy are modeled by equation 22:

$$f(t) \propto I_p(t), \quad f(t) \propto W_p(t) \quad (22)$$

Where $I_p(t)$ is the plasma current (A) and $W_p(t)$ is the macroscopic stored energy (J).

2.4 Signal Processing for MHD Mode Characterization

2.4.1 Time-Frequency Analysis

Spectral resolution Δf (Hz) for standard Fast Fourier Transform (FFT) windows is defined in equation 23 [16]:

$$\Delta f = \frac{1}{T} \quad (23)$$

Where T is the window duration (s). For non-stationary events, Fourier analysis fails to capture rapid transitions [17]. This is why, the Continuous Wavelet Transform, which uses

the Morlet wavelet, is chosen, due to its fine temporal resolution at high frequencies, defined in equation 24 [18]:

$$\psi(t) = \pi^{-1/4} e^{i\omega_0 t} e^{-\frac{t^2}{2}} \quad (24)$$

Where $\psi(t)$ is the Morlet wavelet function and ω_0 is the central frequency parameter (rad/s). Dominant modes must exceed a 3σ noise baseline calculated over a quiescent pre-plasma phase, defined in equation 25 [19], and exhibit global spatial coherence to distinguish modes from single-channel noise [20, 21], defined in equation 26:

$$P(f, t) \geq \mu_{\text{noise}}(f) + 3\sigma_{\text{noise}}(f) \quad (25)$$

Where $P(f, t)$ is the power spectral density (V^2/Hz), f is the frequency (Hz), $\mu_{\text{noise}}(f)$ is the spectral mean of the noise baseline (V^2/Hz), and $\sigma_{\text{noise}}(f)$ is the standard deviation of the noise baseline (V^2/Hz).

$$\gamma_{xy}^2(f) = \frac{|P_{xy}(f)|^2}{P_{xx}(f)P_{yy}(f)} \quad (26)$$

Where $\gamma_{xy}^2(f)$ is the magnitude-squared spatial coherence, $P_{xy}(f)$ is the cross-spectral density (V^2/Hz), and $P_{xx}(f)$ and $P_{yy}(f)$ are the auto-spectral densities (V^2/Hz).

2.4.2 Toroidal Mode Identification & Radial Localization

Toroidal mode number is obtained via cross-spectral phase at frequency f_0 , defined in equation 27 [22, 23]:

$$\Delta\phi_{ij}^{\text{meas}} = \arg [P_{x_i x_j}(f_0)] \quad (27)$$

Where $\Delta\phi_{ij}^{\text{meas}}$ is the measured cross-spectral phase (rad), f_0 is the peak mode frequency (Hz), and x_i and x_j are the respective probe signals (V). The theoretical spatial phase shift is given by equation 28 [24, 25]:

$$\Delta\phi_{ij}^{\text{theo}}(n) = (n\Delta\varphi_{ij} + \pi) \pmod{2\pi} - \pi \quad (28)$$

Where $\Delta\phi_{ij}^{\text{theo}}(n)$ is the theoretical spatial phase shift (rad) and $\Delta\varphi_{ij}$ is the geometric toroidal separation angle (rad). The best-fit n minimizes the root mean square (RMS) phase error, defined in equation 29, incorporating coplanar selection rules to avoid spatial aliasing [26]:

$$\text{RMSE}(n) = \sqrt{\frac{1}{N_{\text{pairs}}} \sum_{(i,j)} (\Delta\phi_{ij}^{\text{meas}} - \Delta\phi_{ij}^{\text{theo}}(n))^2} \quad (29)$$

Where $\text{RMSE}(n)$ is the RMS error (rad) and N_{pairs} is the number of evaluated probe pairs. Radial localization utilizes ECE radial temperature fluctuation profiles, extracted as shown in equation 30, mapping the perturbation peak via cross-coherence with Mirnov signals [27]:

$$\tilde{T}_e(r) = \text{std}(\text{filt}(y_{\text{ECE}})) \quad (30)$$

Where $\tilde{T}_e(r)$ is the RMS electron temperature fluctuation (K), r is the radial coordinate (m), and y_{ECE} is the raw Electron Cyclotron Emission signal (V).

2.5 Signal Conditioning and Correlation Analysis

2.5.1 Instantaneous Parameters & Envelope Dynamics

Magnetic signals are conditioned using a Bessel bandpass filter, selected for its maximally flat group delay to prevent phase distortion [28]. The analytic signal is constructed via the Hilbert transform to extract unwrapped phase and instantaneous amplitude [29]. Prior to correlation, data also undergoes low-pass filtering and anti-aliasing decimation to strictly respect the temporal Nyquist limit [17, 29].

Wave-packet beating creates envelope modulation, defined in equation 31:

$$f_{\text{mod}} = f_0 - f_1 \quad (31)$$

Where f_{mod} is the envelope modulation frequency (Hz) and f_1 is the lower carrier frequency (Hz). Carrier-envelope self-coupling coherence is evaluated via equation 32 [22]:

$$\gamma_{\tilde{B},A}^2(f) = \frac{|P_{\tilde{B}A}(f)|^2}{P_{\tilde{B}\tilde{B}}(f)P_{AA}(f)} \quad (32)$$

Where $\gamma_{\tilde{B},A}^2(f)$ is the self-coupling coherence, $P_{\tilde{B}A}(f)$ is the cross-spectral density between the carrier and envelope (V^2/Hz), $P_{\tilde{B}\tilde{B}}(f)$ is the carrier auto-spectral density (V^2/Hz), and $P_{AA}(f)$ is the envelope auto-spectral density (V^2/Hz). This must exceed the statistical noise floor formulated in equation 33 [30]:

$$\gamma_{95}^2 = 1 - (0.05)^{1/(N_{\text{seg}}-1)} \quad (33)$$

Where γ_{95}^2 is the 95 percent statistical noise floor and N_{seg} is the number of averaging segments.

2.5.2 Spatial Harmonic Decomposition & Poloidal Phase

To resolve the poloidal structure across non-uniform probes and prevent spatial aliasing [31], the spatial non-uniform discrete Fourier transform (NUDFT) is computed via equation 34 [32]:

$$S_k = \frac{1}{N_{\text{probes}}} \sum_{j=1}^{N_{\text{probes}}} S_j e^{-ik\theta_j} \quad (34)$$

Where S_k is the spatial spectral component at wavenumber k (V), k is the discrete poloidal wavenumber, N_{probes} is the total number of probes, S_j is the complex spectral component of probe j (V), and θ_j is the poloidal angle of probe j (rad). Cross-spectral poloidal phase relative to a reference probe is extracted using equation 35 [23]:

$$\phi_j = \arg[\langle S_j(f)S_{\text{ref}}^*(f) \rangle] - \phi_{\text{ref}} \quad (35)$$

Where ϕ_j is the cross-spectral poloidal phase of probe j (rad), S_{ref} is the complex spectral component of the reference probe (V), and ϕ_{ref} is the phase of the reference probe (rad).

This is subject to the Bendat-Piersol metrological phase uncertainty established in equation 36 [22]:

$$\sigma_{\phi,j} = \sqrt{\frac{1 - \gamma_j^2}{2\gamma_j^2 N_{\text{seg}}}} \quad (36)$$

Where $\sigma_{\phi,j}$ is the metrological phase uncertainty (rad) and γ_j^2 is the coherence of probe j with the reference probe. A phase good fit is verified by the Mardia-Jupp circular correlation coefficient, calculated via equation 37 [33]:

$$r_{\text{circ}}(m) = \frac{1}{N} \sqrt{\left(\sum \cos(\phi_j - m\theta_j)\right)^2 + \left(\sum \sin(\phi_j - m\theta_j)\right)^2} \quad (37)$$

Where $r_{\text{circ}}(m)$ is the circular correlation coefficient and N is the sample size of probes. Valid poloidal structures strictly require $r_{\text{circ}} \geq 0.70$ and a mean absolute phase error $|\overline{\Delta\phi}| \leq 45^\circ$, evaluated after correcting known hardware polarity inversions.

2.5.3 Correlation and Significance

Lagged cross-correlation identifies dynamic heating response delays [34], defined in equation 38:

$$R_{xy}(\tau) = \frac{1}{N} \sum_t \left(\frac{x(t) - \bar{x}}{\sigma_x} \right) \left(\frac{y(t + \tau) - \bar{y}}{\sigma_y} \right) \quad (38)$$

Where $R_{xy}(\tau)$ is the lagged cross-correlation, τ is the time lag (s), $x(t)$ and $y(t)$ are the time-series signals (V), $\bar{x}$ and $\bar{y}$ are their respective sample means (V), and σ_x and σ_y are their sample standard deviations (V). These empirical delays are benchmarked against the fast-ion slowing-down timescale defined in equation 39 [35]:

$$\tau_s \approx 6.27 \times 10^8 \frac{A_b T_e^{3/2}}{Z_b^2 n_e \ln \Lambda} \quad (39)$$

Where τ_s is the slowing-down timescale (s), A_b is the beam ion mass number, Z_b is the beam ion charge number, and $\ln \Lambda$ is the Coulomb logarithm [7], for this specific empirical constant, T_e is strictly evaluated in eV, as it is common in plasma physics and n_e in cm^{-3} . Effective sample size is constrained by autocorrelation using equation 40 [36]:

$$N_{\text{eff}} = \frac{N}{1 + 2 \sum_{k=1}^M \rho(k)} \quad (40)$$

Where N_{eff} is the effective sample size, M is the maximum evaluated lag, and $\rho(k)$ is the sample autocorrelation at lag k . Significance testing utilizes the t -statistic in equation 41 [36] and partial correlation in equation 42 to isolate physical coupling from operational artifacts, evaluated alongside False Discovery Rate corrections [37]:

$$t = r \sqrt{\frac{N_{\text{eff}} - 2 - n_{\text{control}}}{1 - r^2}} \quad (41)$$

Where t is the t -statistic, r is the Pearson correlation coefficient, and n_{control} is the number of control variables.

$$r_{xy \cdot z} = \frac{r_{xy} - r_{xz}r_{yz}}{\sqrt{(1 - r_{xz}^2)(1 - r_{yz}^2)}} \quad (42)$$

Where $r_{xy \cdot z}$ is the partial correlation coefficient between x and y controlling for z , and r_{xy} , r_{xz} , and r_{yz} are the respective pairwise Pearson correlation coefficients.

2.6 Higher-Order Spectral Analysis for Nonlinear Wave Coupling

2.6.1 Bicoherence & Coupling Domains

Quadratic nonlinear three-wave coupling is detected via the auto-bispectrum defined in equation 43, measuring interactions that satisfy resonance matching [16, 38, 39]:

$$B(f_1, f_2) = \langle X(f_1)X(f_2)X^*(f_1 \pm f_2) \rangle \quad (43)$$

Where $B(f_1, f_2)$ is the auto-bispectrum (V^3/Hz^2), f_1 and f_2 are the interacting frequencies (Hz), and $X(f)$ is the Fourier transform of the signal (V/Hz). Normalized phase-locking is quantified by the squared auto-bicoherence in equation 44 [16, 38]:

$$b^2(f_1, f_2) = \frac{|\langle X(f_1)X(f_2)X^*(f_1 \pm f_2) \rangle|^2}{\langle |X(f_1)X(f_2)|^2 \rangle \langle |X(f_1 \pm f_2)|^2 \rangle} \quad (44)$$

Where $b^2(f_1, f_2)$ is the squared auto-bicoherence. Significant coupling (typically $b^2 > 0.5$ [21]) must exceed the statistical noise threshold formulated in equation 45 to reject spurious background noise [38, 39]:

$$b_\alpha^2 = -\frac{\ln(\alpha)}{N_{\text{ensemble}}} \quad (45)$$

Where b_α^2 is the statistical significance threshold, α is the significance level, and N_{ensemble} is the number of independent ensembles. To prevent redundant triad integration due to permutation symmetries, the spectra are mathematically partitioned into sum coupling (up-conversion) and difference coupling (parametric decay) regions, defined in equations 46 and 47:

$$f_1 + f_2 = f_3 \quad (f_1 > 0, f_2 > 0) \quad (46)$$

$$f_1 - f_2 = f_3 \quad (f_1 > 0, f_2 < 0) \quad (47)$$

Where f_3 is the resultant coupled frequency (Hz).

2.6.2 Bias Scaling & Window Optimization

Because finite-sample bicoherence exhibits an intrinsic positive bias ($E[\hat{b}^2] \approx 1/N$), physical triads are verified through the asymptotic scaling model shown in equation 48, strictly requiring a non-zero intercept $b_\infty^2 > 0$ [40]:

$$\hat{b}^2(N) = b_\infty^2 + \frac{C}{N} \quad (48)$$

Where $\hat{b}^2(N)$ is the finite-sample squared bicoherence, b_∞^2 is the asymptotic true bicoherence, and C is a scaling constant. Coupled triads geometrically cluster along characteristic locations: subharmonic decay ($f_2 = -0.5f_1$), primary pump decay ($f_1 = f_{\text{pump}}$), and harmonic doubling ($f_1 = f_2$). Valid triads require adaptive temporal stationarity (minimizing instantaneous $|df/dt|$) and linear PSD peak correspondence ($\Delta f \leq 1.5$ bins) [40].

2.6.3 H-mode Burst-Window Detection and Validation

To evaluate degraded transport, elevated coupling is cross-referenced against H-mode burst windows [5, 6]. A hysteresis algorithm triggers onset at $W_p \geq \mu_{\text{base}} + k_{\text{on}}\sigma_{\text{base}}$ and termination at $W_p \leq \mu_{\text{base}} + k_{\text{off}}\sigma_{\text{base}}$. The statistical exclusivity of burst bicoherence is evaluated against the quiescent phase using a non-parametric Kruskal-Wallis test, visible on the equation 49 [36, 37].

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1) \quad (49)$$

Where H is the Kruskal-Wallis test statistic, N is the total number of observations across all groups, k is the number of groups compared, n_i is the number of observations in group i , and R_i is the sum of ranks for group i .

This is followed by post-hoc Mann-Whitney U tests to identify specific pairwise differences, calculated via equation 50:

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1 \quad (50)$$

Where U_1 is the Mann-Whitney U test statistic for group 1, n_1 and n_2 are the sample sizes for group 1 and group 2, and R_1 is the sum of the ranks for group 1.

To control the family-wise error rate across multiple comparisons, a Bonferroni correction is applied to the significance level, as shown in equation 51:

$$\alpha_{\text{adj}} = \frac{\alpha}{m} \quad (51)$$

Where α_{adj} is the adjusted significance level, α is the original target significance level, such as 0.05, and m is the total number of independent hypotheses or pairwise comparisons evaluated.

Physical validation demands cross-diagnostic consistency, matching Mirnov triads to Langmuir electrostatic measurements ($\Delta f \leq 10$ kHz) [5]. Spatial false-positive rejection applies the effective number of independent channels, calculated in equation 52 [41, 42]:

$$M_{\text{eff}} = \sum_{i=1}^{N_{\text{tips}}} \min(\lambda_i, 1) \quad (52)$$

Where M_{eff} is the effective number of independent channels, N_{tips} is the number of probe tips, and λ_i is the i -th eigenvalue of the correlation matrix. Capping the family-wise spatial

chance probability defined in equation 53:

$$P_{\text{chance}} = 1 - \left(1 - \frac{4(\Delta f)^2}{f_{1,\text{max}} f_{2,\text{max}}} \right)^{M_{\text{eff}} \cdot N_{\text{MP}}} \quad (53)$$

Where P_{chance} is the family-wise spatial chance probability, $f_{1,\text{max}}$ and $f_{2,\text{max}}$ are the maximum search frequencies (Hz), and N_{MP} is the number of evaluated Mirnov channels. Final confirmation establishes that elevated three-wave coupling coincides with a degraded rate of stored energy change ($dW_p/dt \leq 0$ kJ/s), mediating enhanced turbulent transport [38, 39].

3 Methodology

Due to the Heliotron J experimental campaigns nature, the parameters are continuously being changed, resulting in different heating behaviors and instrumentation being off for some shots. The unique configuration of the device ($L = 1, M = 4$ helical-axis) allows for high operational flexibility, enabling independent control over the rotational transform, magnetic well depth, and bumpy field components. This flexibility, while beneficial for exploring diverse confinement regimes, fundamentally alters the energetic particle confinement and MHD stability landscape from shot to shot if changed.

Consequently, the shot selection required a rigorous filtering process to ensure consistent magnetic topologies and heating schemes were compared. These shots are available at the link presented in the annex 1 Also, a visually diagnosis for the Mirnov and frequency behavior was made, ensuring that the analysis was being executed on well behaved data that can later be extrapolated to more complex experimental results. This visual screening was critical to reject discharges where the magnetic signals were severely corrupted by operational artifacts, such as power supply switching noise, external RF pickup, or atypical plasma termination events. On the figure 2, the methodology pipeline is presented, being extended on the posterior subsections. The GitHub repository with the python codes used in order to execute this methodology is available at the appendix A.

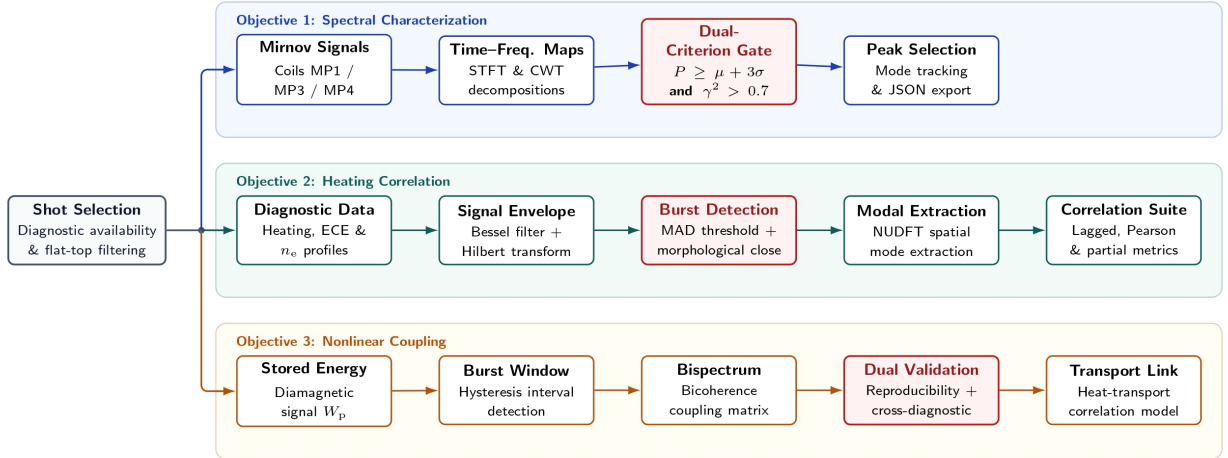


Figure 2: Methodology pipeline. Source: Own elaboration.

3.1 First Specific Objective

This objective centers in the characterizing of the spectral properties and temporal evolution of the MHD oscillations in the Heliotron J device [10, 11]. For this matter, a Python pipeline applies a rigorous dual-criterion validation to isolate physical modes from background noise that will be expanded next.

3.1.1 Data Acquisition and Preprocessing

Magnetic fluctuation data from Mirnov coils MP1, MP3, and MP4 were imported via the `turnelib` library from the `jpack` library, available in the annex 2. MP1 served as the spectral reference, while MP3 and MP4 provided spatial cross-validation. Channel MP2 was specifically excluded from this configuration due to its non-zero poloidal angle, which would introduce confounding phase artifacts into the purely toroidal mode analysis, compromising the validity of the results [25, 26].

3.1.2 Time-Frequency Analysis

Two complementary representations of the MP1 signal captured stationary and transient dynamics:

- STFT: A windowed Fast Fourier Transform with the `jpack` package called `libana_signal`, available in the annex 2 using a Hann window generated a Power Spectral Density (PSD) spectrogram presented in the equation 23 [16], dynamically scaled for a constant 500 Hz resolution.
- CWT: A complex Morlet wavelet ($\omega_0 = 6$) based in equation 24 [17, 18] resolved fast transients. Time support scaled to ± 5 standard deviations across 150 frequency bins (10 kHz to 150 kHz).

3.1.3 Dual-Criterion Mode Identification

For a mode to be a confirmed physical mode, it requires simultaneous statistical and spatial validation:

- Statistical Dominance: A baseline 3σ threshold, as seen in the equation Eq. 25 [19] was calibrated using a 0 to 25 ms pre-plasma window. Smoothed spectral power (4 ms rolling average) had to continuously exceed this threshold for ≥ 2 ms.
- Spatial Coherence: Welch's method (100 ensembles) evaluated magnitude-squared coherence (γ^2), visible in the equation 26 [20, 21] between MP1-MP3, MP1-MP4, and MP3-MP4. The convention for valid modes requires $\gamma^2 > 0.7$ on at least one pair.

3.1.4 Peak Selection and Export

Finally, the prominent peaks (15 kHz to 100 kHz) that passed the criteria were isolated, separating the dominant only against the dominant and coherent. Mode properties, such as

frequency, amplitude, active intervals were exported to a .JSON file, alongside visual diagnostic summaries such as a time dependent 3σ spectrogram, the Morlet Wavelet Transform time dependency, and the coherence distributions for all the frequencies.

3.2 Second Specific Objective

The second objective mission is to evaluate correlations between EPs and plasma heating (NBI and ECH) using the magnetic fluctuation frequencies [3, 4, 34], more detail is presented next.

3.2.1 Signal Conditioning and Active Burst Detection

Heating, ECE, and density signals were linearly interpolated onto the Mirnov time base. The complete signal-conditioning and dynamic parameter extraction sequence is summarized in the figure 3 and is expanded on the following subsections, providing a deeper submersion on the data treatment.

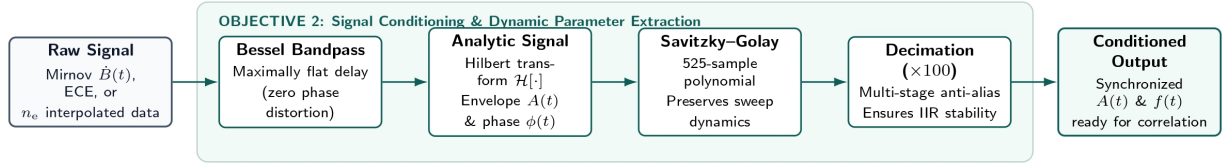


Figure 3: Signal-conditioning and parameter-extraction chain for Objective 2. Source: Own elaboration.

Starting with a Bessel bandpass filter and a Hilbert transform that extracted the mode’s instantaneous amplitude and phase [28, 29]. Then, to make sure that the physical frequency sweeps are not artificially flattened, the Savitzky-Golay filter utilizes a statically optimized window size of 525 samples, a value confirmed via offline sensitivity sweeps to best preserve the mode’s instantaneous frequency dynamics.

Analyses were restricted to a dynamically thresholded active window. This threshold is established using the full-trace median plus 6.0 times the standard-deviation-equivalent Median Absolute Deviation ($1.4826 \times \text{MAD}$) to accurately represent the heated baseline noise floor. A binary closing morphological operation, bridging gaps up to 3.0 ms, is subsequently applied to isolate the single largest contiguous mode-active burst, rejecting noise spikes shorter than 10.0 ms. Furthermore, prior to correlation analysis, the envelope and frequency arrays undergo a cascaded anti-aliasing decimation, with a total factor of 100, utilizing iterative intermediate downsampling steps to mathematically maintain low-pass Infinite Impulse Response (IIR) filter stability while respecting the temporal Nyquist limit [17, 29].

3.2.2 Modal Structure Extraction

A non-uniform discrete Fourier transform (NUDFT) across coils PMP1–PMP14 was used in order to extract the poloidal mode number (m) from the analyzed mode at this shot (Eq. 34, 35) [31]. Voltages were converted to T/s, and inverted wirings (PMP1–PMP4) were corrected, from the original code provided by the Heliotron J team, and available in the annex 3. Search boundaries spanned $m = \pm 6$, automatically expanding to ± 20 if peak-pinned. Inter-probe coherence was gated by a flat-frequency sub-window algorithm anchored to the Objective 1 active intervals. A sliding 8.0 ms window located the core with the lowest local slope $|df/dt|$, growing within a 50 % tolerance to evaluate coherence strictly over the non-chirping plateau.

3.2.3 Correlation with Heating Systems

Mode parameters and heating profiles were synchronized using statistical metrics:

- Lagged Cross-Correlation: Scanned ± 20.0 ms for amplitude, ± 40.0 ms for frequency, and ± 60.0 ms for core proxies in order to evaluate hysteresis, as stated in equation 38 [34].
- Pearson Correlation: Required $|r| > 0.7$ and $p < 0.05$, adjusting effective sample size (N_{eff}) for serial autocorrelation, like in equations 40 and 41 [36].
- Partial Correlation: Rejected shared-step artifacts (e.g., ECH turn-off), verified via binarization threshold sweeps (30 % to 70 %), this is based in equation 42 [37].
- Correction: Applied Bonferroni and the Benjamini-Hochberg procedure for the False Discovery Rate FDR-BH adjustments, systematically excluding low-power tests ($N_{\text{eff}} < 5$) [43, 44].

3.2.4 Energetic-Particle Distribution and Alfvén Scaling

Temporal delays between mode dynamics and core ECE proxies were evaluated against the Spitzer electron-drag timescale (τ_s) presented in equation 39 [35]. ECE channels were selected by highest zero-lag correlation to ECH power, strictly filtering ADC-saturated channels (e.g., > 20 static samples). Frequency sweeping was validated against Alfvén velocity scaling, displayed in equations 13 and 17 [7, 15] using a 1.25 T field and median-filtered ($n = 5$) electron density, interpolating over transient ECH glitches. Invariance was verified across calibration windows from 150.0 ms to 285.0 ms.

3.3 Third Specific Objective

This final specific objective characterizes the low-frequency H-mode fluctuations by mapping nonlinear coupling events associated with edge turbulence and heat transport [5, 6], as presented next.

3.3.1 H-mode Burst-Window Detection and Sub-Window Refinement

The stored energy signal (W_p) partitioned the discharge into burst ($k_{\text{on}} = 3.0$) and back-transition ($k_{\text{off}} = 1.5$) windows [5], requiring a 10 ms minimum duration. To define the quiescent and control windows, the algorithm searched outward from the burst edges in 2.0 ms increments, starting beyond a 5.0 ms boundary margin, to find the nearest genuinely flat, sub-threshold 25.0 ms segments. The acceptable flatness criteria was strictly self-calibrated to the discharge’s empirical background, requiring the segment’s variance score to fall below 1.5 times the median (50th percentile) spread of equivalently sized background windows. Bicoherence limits were tightened via an 8.0 ms flat-frequency sub-window applied to MP1 (Bessel-filtered 5.0–100.0 kHz), grown within a 50 % tolerance until chirping limits forced iteration to meet FFT sample floors ($N_{\text{fft}} = 1024$, $N_{\text{ensemble}} = 30$).

3.3.2 Bicoherence Calculation and Peak Identification

Squared auto-bicoherence $b^2(f_1, f_2)$ evaluated Sum and Difference interactions, presented in equations 43, 44, 46 and 47 [38, 39]. Significance required surpassing $b^2 > 0.5$ [21] and a 95 % statistical noise threshold ($b_{95}^2 = -\ln(0.05)/N_{\text{ensemble}}$) stated in equation 45 [38, 39]. Searches were restricted below 100.0 kHz with a 5.0 kHz minimum floor. Peaks identified via a 2D maximum filter (5.0 kHz) were subjected to boundary (10.0 kHz) and inner-core stability checks. Triads were confirmed within 1.5 bins of genuine PSD peaks. An RMS amplitude check flagged array channels deviating $5\times$ from the median.

3.3.3 Validation Criteria

Statistical superiority of burst-window bicoherence was verified via Kruskal-Wallis and Mann-Whitney U tests ($\alpha = 0.025$) [36, 37]. Candidate couplings required strict double-validation:

- Reproducibility: Leading peaks ($b^2 > 0.5$) must replicate within ± 10.0 kHz in an independent validation discharge.
- Cross-Diagnostic Consistency: Magnetic signatures must concurrently appear in the divertor Langmuir array (excluding non-physical bias/trigger pins). Tip couplings were gated by a multiple-comparisons check ($p < 0.05$) based on the array’s effective independent channels, based in equations 52 and 53 [41, 42].

3.3.4 Heat Transport Correlation

Elevated burst-phase bicoherence was ultimately validated against periods of degraded or negative temporal stored energy derivatives ($\langle dW_p/dt \rangle$), confirming three-wave coupling mediation of enhanced turbulent transport [38, 39].

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Appendix A: GitHub repository

The code made for the methodology is uploaded at a following GitHub repository, and it is available in the following [link](#).

Annex 1: Experimental Data

The experimental data for all of the shots analyzed is available at the following [link](#).

Annex 2: Jpack Python Library

The Jpack python library, created by the contact for this thesis, Ph.D. Prof. Shigeru Inagaki is available at the following [link](#)

Annex 3: Poloidal mode number extraction

The following code, presented on the listing 1 was created by the contact for this thesis, Ph.D. Prof. Shigeru Inagaki, in order to calculate the Auto Power Spectrum for a time-series-data and is not available in the annex 2, but was used and adapted for the present work.

Listing 1: Auto power spectrum PMP.

```
1  #!/usr/bin/env python
2  """
3      This application calculates auto power spectrum of a time-series-data.
4      The time-series-data should be given as "edf" format.
5      Outputs are results of
6          1. running FFT
7          2. Auto Power Spectrum which is ensemble averaged over a specified
   ↪ time period.
8          3. AUto Bi-Coherence which is ensemble averaged over a specified time
   ↪ period.
9          4. Total Bi-Coherence
10
11  Usage:
12  anaaspect [-h] [--version] [-f filename] [-t id] -r fres
13
14  optional arguments:
15      -h, --help                Show this help message and exit
16      --version                Show program's version number and exit
17      -f filename, --file filename
18                               Set an edf-formatted file to be read
19      -t id, --target id       Set variables IDs to be plotted (default 1)
20      -r fres, --resolution fres
21                               Set frequency resolution in [Hz]
22
23  Example:
24      python anaaspect.py -f ../expdata/FY2022/MP3@83489.edf -r 500
25
26  Dependence
27  -----
28  turnelib.py
29  libana_signal.py
30  matplotliblibutil.py
31
32  Status
33  -----
34  Version 0.9.3.2
35
36  Authour
37  -----
38  Shigeru Inagaki
39  Institute of Advanced Energy, Kyoto University
40  s-inagaki@iae.kyoto-u.ac.jp
41
42  Revision History
43  -----
44  [29-Nov-2022] Creation                                ver 0.9.3
45  [19-Dec-2022] Spectrum is log-log plotted            ver 0.9.3.1
```

```

45 [19-Dec-2022] matplotlibutil has been updated ver 0.9.3.2
46 Copyright
47 -----
48 2022 Shigeru Inagaki (s-inagaki@iae.kyoto-u.ac.jp)
49 Released under the MIT, BSD, and GPL Licenses.
50
51 """
52
53 import os
54 import numpy as np
55 import scipy.signal as dsp
56 import scipy.fftpack as fft
57 import libana_signal as LAS
58 import turnelib as TE
59 import hplib as HJ
60 import matplotlib
61 import matplotlib.pyplot as plt
62 import matplotlibutil as MPU
63 from matplotlib.widgets import TextBox, Button
64
65 matplotlib.axes.Axes.add_trace = MPU.add_trace
66 matplotlib.axes.Axes.add_errorbar = MPU.add_errorbar
67 matplotlib.axes.Axes.matview = MPU.matview
68 matplotlib.figure.Figure.make_subplots = MPU.make_subplots
69
70 def getdata_online(shotnum, diagnames, convmode, offsmode, offs1, offs2,
    ↪ reverse):
71     diag = HJ.hjb()
72     ndiag = len(diagnames)
73     for i in range(ndiag):
74         if i == 0 :
75             raw, time = diag.get(shotnum, diagnames[i])
76             data = np.zeros([ndiag, diag.dlen])
77         else:
78             raw, _ = diag.get(shotnum, diagnames[i])
79             data[i,:] = HJ.convert(raw, time, convmode, diag.scale, offsmode,
    ↪ offs1, offs2, reverse)
80     return data, time
81
82 def fiscalyear(shotnum):
83     hdiskdir, ddir = HJ.getdatadir(shotnum)
84     i = ddir.find('/')
85     date = ddir[:i]
86     yy = int(date[:1])
87     mm = int(date[2:3])
88     if mm < 3 :
89         fiscal = yy - 1
90     else :
91         fiscal = yy
92     return fiscal
93
94 def getdata_offline_old(filename):
95     diag = TE.edf()
96     raw = diag.load(filename)

```

```

97     time = raw[:,0]
98     data = raw[:,1:].transpose()
99     if diag.DimUnit[0] == 'ms':
100         time = time/1000.0
101     return data, time
102
103 def getdata_offline(shotnum, diagnames, dir=None):
104     diag = TE.edf()
105     ndiag = len(diagnames)
106     for i in range(ndiag):
107         filename = diagnames[i]+'@{0:05}.edf'.format(shotnum)
108         if dir is not None:
109             filename = dir + '/' + filename
110         raw = diag.load(filename)
111         if i == 0 :
112             data = np.zeros([ndiag, diag.DimSize[0]])
113             time = raw[:,0]
114         else :
115             data[i,:] = raw[:,1].transpose()
116         if diag.DimUnit[0] == 'ms':
117             time = time/1000.0
118     return data, time
119
120 def getPMPs(shotnum, online=True, dir=None):
121     mpns = 4.5e-3 # coil parameter (height x width x
122     ↪ turns) for PMP
123     mpgain = 2. # amplifier gain, (usually seted
124     ↪ to x 10) for PMP
125     PMPnames = ['PMP1', 'PMP2', 'PMP3', 'PMP4', 'PMP5', 'PMP6', 'PMP7',
126                 'PMP8', 'PMP9', 'PMP10', 'PMP11', 'PMP12', 'PMP13', 'PMP14']
127
128     if (shotnum >= 67829) and (shotnum <= 68901):
129         PMPnames[13] = 'DUM-PMP14'
130
131     if online :
132         data, time = getdata_online(shotnum, PMPnames, 'V', 'C', 0, 0, False)
133         data = data/(mpgain*mpns) # change voltage (V) to physical
134         ↪ value (T/s)
135     else :
136         # filename = 'PMP@{0:05}.edf'.format(shotnum)
137         # data, time = getdata_offline(filename)
138         data, time = getdata_offline(shotnum,PMPnames, dir)
139     return data, time
140
141 def getMPs(shotnum, online=True, dir = None):
142     MPnames = ['MP1', 'MP3']
143     target = [0,2]
144     if online :
145         data, time = getdata_online(shotnum, MPnames, 'V', 'C', 0, 0, False)
146         # change voltage (V) to physical value (T/s)
147         # NOTE: MPs were replaced at before 11th campaign
148         fyear = fiscalyear(shotnum)
149         if (fyear >= 11) :
150             data = data/(30.*15.*1.e-6*20.*10.)

```

```

148     else :
149         data = data/(25.*25.*1.e-6*20.*10.)
150     # check polarity of MP
151     # Mirnov coils polarity was check on Apr. 22, 2014 (syama)
152     # 4 May, 2014 : Polarity of MP for FY11, 12 and 13 are correct
153     if (fyear == 10) :
154         data[0,:] = -data[0,:]
155     return data, time
156 else :
157     # filename = 'MP@{0:05}.edf'.format(shotnum)
158     # data, time = getdata_offline(filename)
159     data, time = getdata_offline(shotnum, MPnames, dir)
160     return data, time
161
162 def checkAutoPower(sig, t, fs, nfft, title):
163     f, tave, Pyy = LAS.running(sig, t, dt=1/fs, nfft=nfft, noverlap = nfft//2,
164     ↪ detrend='constant')
165     tt, ff = np.meshgrid(tave,f)
166
167     page = MPU.mypage()
168     plots = page.make_subplots(1,1)
169     Pmax = TE.maxval(Pyy)
170     Pmin = TE.minval(Pyy)
171     cont = plots[0].contourf(tt*1000, ff/1000, np.log10(Pyy), cmap='jet',)
172     cb = plt.colorbar(cont) #ticks=[0.0001,0.001,0.01,0.1,1])
173     cb.set_label('log10 PSD (V**2/Hz)')
174     plots[0].set_ylabel('f (kHz)')
175     plots[0].set_xlabel('t (ms)')
176     plots[0].set_title(title, loc='right', fontdict={'fontfamily':'monospace',
177     ↪ 'fontsize':12,})
178     ymin, ymax = plots[0].get_ylim()
179     for t12 in tranges:
180         plots[0].vlines(t12[0],ymin,ymax,linestyle='dashed')
181         plots[0].vlines(t12[1],ymin,ymax,linestyle='dashed')
182     plt.show()
183
184 def runningCSD(sigs, t, ref, fs=None, nfft=256, noverlap=None, nensemble=1,
185     ↪ nstep=None, window='hann', detrend='constant'):
186     if noverlap is None :
187         noverlap = nfft // 2
188     ncalc = nensemble*nfft -noverlap*(nensemble-1)
189     if nstep is None :
190         nstep = ncalc - noverlap
191     ntime = len(t) // ncalc
192     tave = np.zeros(ntime)
193     nch = sigs.shape[0]
194     xcoh2 = np.zeros([nch, nfft, ntime])
195     xphase = np.zeros([nch, nfft, ntime])
196     for i in range(ntime):
197         istart = i*nstep
198         iend = istart + ncalc
199         f, Pxy = dsp.csd(sigs[:,istart:iend], ref[istart:iend], fs=fs, window=
200     ↪ window, nperseg=nfft, noverlap=noverlap,
201         nfft=None, detrend=detrend, return_onesided=False,

```

```

198     ↪ scaling='density', axis=1)
    _, Pyy = dsp.welch(sigs[:,istart:iend], fs=fs, window=window, nperseg=
199     ↪ nfft, noverlap=noverlap, nfft=None, detrend=detrend,
        return_onesided=False, scaling='density', axis=1)
200     _, Pxx = dsp.welch(ref[istart:iend], fs=fs, window=window, nperseg=
    ↪ nfft, noverlap=noverlap, nfft=None, detrend=detrend,
        return_onesided=False, scaling='density', axis=1)
201     tave[i] = np.mean(t[istart:iend])
202     Pxx = fft.fftshift(Pxx)
203     Pyy = fft.fftshift(Pyy, axes=1)
204     Pxy = fft.fftshift(Pxy, axes=1)
205     pwr = (Pxy*Pxy.conj()).real
206     for j in range(nch):
207         xcoh2[j,:,i] = pwr[j,:]/(Pyy[j,:]*Pxx[:])
208     for k in range(nfft):
209         xphase[:,k,i] = np.unwrap(np.angle(Pxy[:,k]))/np.pi
210     f = fft.fftshift(f)
211     return xcoh2, xphase, f, tave
212
213
214 def runningFFT(sig, fs, nfft=256, noverlap=None, window='hann', detrend='
    ↪ constant'):
215     # sig : 1d array or 2d array
216     #     Time series of measurement values. sig[n, :] means temporal
    ↪ evolution of (n-1)th signal.
217     # S : 2d array or 3d array
218     #     S.shape = (nch,nf,nt)
219     if noverlap is None :
220         noverlap = nfft // 2
221     f, tave, S = dsp.spectrogram(sig, fs=fs, window=window, nperseg=nfft,
    ↪ noverlap=noverlap, nfft=nfft, detrend=detrend,
        return_onesided=False, scaling='density',
222     ↪ axis=-1, mode='complex')
223
224     S = fft.fftshift(S, axes = -2)
225     f = fft.fftshift(f)
226     return f, tave, S
227
228 def dftfreqs(mdft, dk):
229     # mdft, dk : int & float
230     #     Parameters specifying the desired frequency grid. Transform will be
231     #     computed at frequencies dk * (-(mdft//2) + arange(mdft))
232     #     mdft --> nfft
233     return dk * np.arange(-(mdft // 2), mdft - (mdft // 2))
234
235 def nudft(x, func, k, sign = -1):
236     # x, func : array_like
237     #     real locations x and complex values c of the points to be transformed
    ↪ .
238     n = len(x)
239     return (1 / n) * np.dot(func, np.exp(sign * 1j * k * x[:,None]))
240
241 #def submit(text):
242 #     ydata = eval(text)
243 #     l.set_ydata(ydata)

```

```

244 # ax.set_ylim(np.min(ydata), np.max(ydata))
245 # plt.draw()
246
247 my_round_int = lambda x: int((x * 2 + 1) // 2)
248
249 if __name__ == "__main__":
250     version = "0.9"
251     timestamp = "2022/10/26"
252
253     myparser = TE.default_parser('anaaspect', 'Spectrum analysis (Auto-Spectrum
↪ )', version, timestamp)
254     myparser.add_argument(
255         '-s',
256         '--shot',
257         action='store',
258         required = True,
259         type=int,
260         default = None,
261         metavar = 'shot',
262         help='Set shotnumber'
263     )
264     myparser.add_argument(
265         '-r',
266         '--resolution',
267         action='store',
268         type=float,
269         default = None,
270         required = False,
271         metavar = 'fres',
272         help='Set frequency resolution in [Hz]'
273     )
274     args = myparser.parse_args()
275
276     shotnum = args.shot
277     ToRadian = np.pi/180.0
278     online = True
279
280     PMPs, time = getPMPs(shotnum, online)
281     MPs, time = getMPs(shotnum, online)
282     fs = 1.0e6
283     nfft = 1000 # width of data window (default: 1000)
284
285     ### Check MP3 and set time-of-interest
286     f, tave, Pyy = LAS.running(MPs[1,:], time, dt=1/fs, nfft=nfft, noverlap =
↪ nfft//2, detrend='constant')
287     tt, ff = np.meshgrid(tave,f)
288
289     title = 'MP3@{0:05}'.format(shotnum)
290     page = MPU.mypage()
291     plots = page.make_subplots(1,1, bottom=0.4)
292     cont = plots[0].contourf(tt*1000, ff/1000, np.log10(Pyy), cmap='jet',)
293     cb = plt.colorbar(cont) #ticks=[0.0001,0.001,0.01,0.1,1])
294     cb.set_label('log10 PSD (V**2/Hz)')
295     plots[0].set_ylabel('f (kHz)')

```

```

296 plots[0].set_xlabel('t (ms)')
297 plots[0].set_title(title, loc='right', fontdict={'fontfamily':'monospace',
↪ 'fontsize':12,})
298
299 tranges = []
300 inputs = []
301 def SetRanges(event):
302     global tranges, inputs, vline1, vline2
303     text = text_box.text
304     d = text.split(',')
305     s = [d[0].strip(),d[1].strip()]
306     v = [float(d[0]), float(d[1])]
307     inputs.append(s)
308     tranges.append(v)
309     ymin, ymax = plots[0].get_ylim()
310     vline1 = plots[0].vlines(v[0],ymin,ymax,linestyle='dashed')
311     vline2 = plots[0].vlines(v[1],ymin,ymax,linestyle='dashed')
312     return
313
314 def Reset(event):
315     vline1.remove()
316     vline2.remove()
317     last = tranges.pop()
318     return
319
320 def MoveOn(event):
321     plt.close()
322     return
323
324 axbox = plt.axes([0.4, 0.2, 0.2, 0.06])
325 text_box = TextBox(axbox, 't1, t2 [ms] ? ', initial='200, 300')
326 axbutton_set = plt.axes([0.15, 0.1, 0.1, 0.06])
327 button_set = Button(axbutton_set, 'Set')
328 button_set.on_clicked(SetRanges)
329 axbutton_reset = plt.axes([0.26, 0.1, 0.1, 0.06])
330 button_reset = Button(axbutton_reset, 'Reset')
331 button_reset.on_clicked(Reset)
332 axbutton_ok = plt.axes([0.37, 0.1, 0.1, 0.06])
333 button_ok = Button(axbutton_ok, 'OK')
334 button_ok.on_clicked(MoveOn)
335 plt.show()
336 print('Times of interest:')
337 print(tranges)
338
339 ### Calcurate 2D FFT by using PMP signals
340 # label (poloidal angle) of each PMP
341 plab = np.array([0., 10., 20., 30., 40., 50., 60., 80., 90., 100., 110.,
↪ 120., 150., 180., 360.])
342 # (CW in poloidal direction)
343 plab = 360. - plab
344 plab = ToRadian*plab[::-1] #Sort ascendingly
345
346 # lmax = 2*np.pi*r here r = 1
347 dk = 1.0 # = 2*np.pi/lmax

```

```

348 lmin = ToRadian*10.0
349 kmax = 180.0/10.0 # = (2*np.pi/lmin)/2
350 nk = len(plab)
351 nk = 2*int(kmax)
352 k = dftfreqs(nk, dk)
353
354 count = 0
355 for t12 in tranges :
356     indx = np.where(np.logical_and(time>=t12[0]/1000, time<=t12[1]/1000))
357     f, tave, Spmp = runningFFT(PMPs[:,indx[0]], fs, nfft, noverlap=nfft
↪ //2, window='hann', detrend='constant')
358     tave = tave + time[indx[0][0]] - 1/fs
359     nf = len(f)
360     nt = len(tave)
361
362     Sappend = np.zeros([1, nf, nt], dtype=complex)
363     Sappend[0,:,:] = Spmp[0,:,:] # 0 deg data is appended as 360 deg
↪ data
364     Spmp = np.append(Spmp, Sappend, axis=0)
365     Spmp = Spmp[::-1,:,:] #Sort ascendingly in the plab axis
366
367     P2d = np.zeros([nk,nf], dtype=float)
368     for i in range(nt):
369         for j in range(nf):
370             Sk = nudft(plab, Spmp[:,j,i], k)
371             P2d[:,j] = P2d[:,j] + (Sk*Sk.conj()).real
372     P2d[:,:] = P2d[:,:]/nt
373
374     ilocp = TE.firstloc(f >= -0.0001)
375     kk, ff = np.meshgrid(k, f[ilocp:])
376     title = 'PMPs@{0:05} t={1}-{2} ms'.format(shotnum,inputs[count][0],
↪ inputs[count][1])
377     page = MPU.mypage()
378     plots = page.make_subplots(1,1)
379     cont = plots[0].contourf(kk, ff/1000, np.log10(P2d[:,ilocp:].transpose
↪ ()), cmap='jet',)
380     cb = plt.colorbar(cont) #ticks=[0.0001,0.001,0.01,0.1,1])
381     cb.set_label('log10 PSD (V*2/Hz/m)')
382     plots[0].set_ylabel('f (kHz)')
383     plots[0].set_xlabel('Poloidal Mode')
384     plots[0].set_title(title, loc='right', fontdict={'fontfamily':'
↪ monospace','fontsize':12,})
385     plt.show()

```